Celia Vallotton

Cell: *(707) 494-3769* | Email: *celiahopes@gmail.com* | Santa Rosa, CA

Education

Santa Rosa Junior College (SRJC) — Santa Rosa, CA
Mechanical Engineering Coursework | Transfer Target: 2028 | GPA 4.0

Relevant Coursework: Calculus I–IV, Engineering Statics, Physics (Mechanics, Waves, Optics, & Thermodynamics, Electricity & Magnetism), MATLAB, Manual Machining, CNC Machining, CAD (Autodesk Inventor), Metallurgy

SKILLS

Software: SolidWorks, Inventor, FeatureCAM, MATLAB, Excel
Technical: CNC Machining & Programming (Haas VF-2, Haas ST-20Y), Manual Machining, 3D printing, Rapid Prototyping, Weld Fixture Design, Tube Notching
Languages: English (Fluent), Spanish (Intermediate)

ENGINEERING EXPERIENCE & PROJECTS

Baja SAE — Chassis Lead | Santa Rosa Junior College
August 2025 – Present

- Led chassis subsystem design for a Baja SAE competition vehicle, developing a rule compliant roll cage while coordinating integration with vehicle subsystems
- Designed and fabricated chassis welding fixtures, developed tube notching processes, and incorporated manufacturability considerations into design
- Led a collaborative team of members from SRJC Society of Women Engineers through the design and fabrication of a full-scale PVC chassis prototype to evaluate driver ergonomics, including seating geometry, pedal positioning, and driver egress
- Conducted SolidWorks Simulation FEA to evaluate chassis loading and stress concentrations from predicted impacts to ensure proper gusset and reinforcement placement

SkillsUSA — Ocean Wave Energy Generator | 3rd Place [California State Conference] | Healdsburg High School
August 2024 – June 2025

- Collaborated on a three-person team to design, prototype, and test an ocean wave energy generator for an engineering design competition
- Sourced materials based on environmental impact, marine exposure, finding water tight solutions, and ensuring cost efficiency
- Utilized rapid prototyping methods, including 3D printing and modular 80/20 aluminum framing, to allow for easy assembly and iterative design modifications
- Successfully tested the system in the ocean, and generated sufficient electricity to illuminate two LEDs

Community STEM Education – Rube Goldberg Program | The Healdsburg School (THS)
November 2025– Present

- Collaborated with a local school to reintroduce hands-on Rube Goldberg engineering projects into our community
- Co-developed and carried out a two-day instructional program for a fourth-grade class covering simple machines, and teaching them how to construct their own Rube Goldberg Machine

LEADERSHIP & PROFESSIONAL EXPERIENCE

Sonoma Millworks — Part Time | Healdsburg, CA
June 2025 – Present

- Assisted in the fabrication and finishing of custom wood products, using table saws, chop saws, belt sanders, drill presses, branders, and finishing products
- Contributed to product development and production of VinoPlank Sprig Vases using reclaimed wood
- Supported showroom organization, product display, and preparation of custom products for retail and wholesale distribution

Society of Women Engineers (SWE) — Vice President | Santa Rosa Junior College
Member: Fall 2025 - Present | Vice President: Fall 2026 - Present

- Coordinated outreach activities and organized opportunities for student engagement in STEM

Coffee Business — Co-Owner | Cequin Coffee Company (www.cequincoffee.com)
2020 – Present

- Coordinated coffee roasting operations, finance management, customer communications, subscription management, inventory procurement, order fulfillment, coffee kiosk construction, and event planning